

Surface Water & Hydrology

FE Environmental — Day 5

Study set • 2026-06-21 • 20 questions • every numeric answer recomputed in Python

Handbook fidelity. Per the NCEES FE Reference Handbook v10.6 (Civil Eng. → Hydrology/Water Resources, p. 296): The Rational Method $Q = CIA$ (US units; note the Handbook's capital I) and the NRCS/SCS rainfall-runoff equations $Q = (P - 0.2S)^2 / (P + 0.8S)$ with $S = 1000/CN - 10$ (inches) are printed verbatim in the Handbook. The following are standard practice but are NOT printed in v10.6 — memorize them: the SI Rational form ($/360$), the SI curve-number form $S = 25400/CN - 254$ (mm), the standalone $I_a = 0.2S$, return period $T = 1/P$, the risk formula $1 - (1 - 1/T)^n$, and the IDF intensity equation.

Q1. [D5-Q01 • MCQ • Rational Method (USCS)]

A watershed has a drainage area of 200 acres, a runoff coefficient $C = 0.65$, and a design rainfall intensity of 2.5 in/hr. Using the Rational Method, the peak runoff rate is most nearly:

Relevant: $Q = CIA$ (USCS: C dimensionless, I in in/hr, A in acres, Q in ft^3/s)

- (A) 200 cfs
- (B) 278 cfs
- (C) **325 cfs** ✓
- (D) 385 cfs

Answer: (C)

$$Q = CIA = 0.65 \times 2.5 \times 200 = 325 \text{ cfs.}$$

Handbook: Surface Water — Rational Method • Ctrl-F: Rational Method

Q2. [D5-Q02 • MCQ • Rational Method (SI)]

A 40-ha urban watershed has a runoff coefficient $C = 0.55$ and a design intensity of 72 mm/hr. Using the SI Rational Method, the peak runoff rate is most nearly:

Relevant: $Q = \frac{CIA}{360}$ (SI: I in mm/hr, A in ha, Q in m^3/s)

- (A) 2.2 m^3/s
- (B) 3.1 m^3/s
- (C) **4.4 m^3/s** ✓
- (D) 5.8 m^3/s

Answer: (C)

$$Q = \frac{0.55 \times 72 \times 40}{360} = \frac{1584}{360} = 4.4 \text{ m}^3/\text{s.}$$

Handbook: Surface Water — Rational Method • Ctrl-F: Rational Method

Q3. [D5-Q03 • MCQ • SCS/NRCS: maximum potential retention S (USCS)]

A watershed has a SCS Curve Number $CN = 80$. The maximum potential retention S (USCS) is most nearly:

Relevant: $S = \frac{1000}{CN} - 10$ (USCS, S in inches)

- (A) 1.5 in
- (B) 2.0 in
- (C) **2.5 in ✓**
- (D) 3.0 in

Answer: (C)

$$S = \frac{1000}{80} - 10 = 12.5 - 10 = 2.5 \text{ in.}$$

Handbook: Surface Water — SCS Runoff • Ctrl-F: Curve Number; SCS

Q4. [D5-Q04 • MCQ • SCS/NRCS: runoff depth (USCS)]

A watershed with $CN = 80$ receives a total storm precipitation $P = 4.0$ in. Using the SCS method, the direct runoff depth Q is most nearly:

Relevant: $S = \frac{1000}{CN} - 10$ $Q = \frac{(P - 0.2S)^2}{P + 0.8S}$ ($P > 0.2S$)

- (A) 1.5 in
- (B) **2.0 in ✓**
- (C) 2.5 in
- (D) 3.0 in

Answer: (B)

$$S = 2.5 \text{ in; } I_a = 0.2(2.5) = 0.5 \text{ in. } P > I_a, \text{ so } Q = \frac{(4.0 - 0.5)^2}{4.0 + 0.8(2.5)} = \frac{12.25}{6.0} = 2.04 \text{ in} \approx 2.0 \text{ in.}$$

Handbook: Surface Water — SCS Runoff • Ctrl-F: Curve Number; SCS

Q5. [D5-Q05 • MCQ • SCS initial abstraction I_a]

For a watershed with $CN = 75$, the SCS initial abstraction I_a (the precipitation that must fall before any runoff occurs) is most nearly:

Relevant: $I_a = 0.2S$, $S = \frac{1000}{CN} - 10$

- (A) 0.33 in
- (B) 0.50 in
- (C) **0.67 in ✓**
- (D) 1.00 in

Answer: (C)

$$S = \frac{1000}{75} - 10 = 3.33 \text{ in}; \quad I_a = 0.2(3.33) = 0.67 \text{ in.}$$

Handbook: Surface Water — SCS Runoff • Ctrl-F: Initial Abstraction; SCS

Q6. [D5-Q06 • MCQ • SCS method: zero-runoff condition]

A watershed has $CN = 75$ ($S = 3.33$ in, $I_a = 0.67$ in). A storm produces only $P = 0.5$ in of precipitation. The direct runoff depth Q from this storm is:

Relevant: If $P \leq I_a = 0.2S$, then $Q = 0$

- (A) **0.00 in ✓**
- (B) 0.08 in
- (C) 0.15 in
- (D) 0.50 in

Answer: (A)

$P = 0.5 \text{ in} < I_a = 0.67 \text{ in}$, so the entire precipitation is captured by initial abstraction and $Q = 0$.

Handbook: Surface Water — SCS Runoff • Ctrl-F: Curve Number; SCS

Q7. [D5-Q07 • MCQ • Return period to annual exceedance probability]

The "50-year storm" has an annual exceedance probability (AEP) of:

Relevant: $P = \frac{1}{T}$

- (A) 1%
- (B) **2% ✓**
- (C) 5%
- (D) 50%

Answer: (B)

$$P = \frac{1}{T} = \frac{1}{50} = 0.02 = 2\%.$$

Handbook: Surface Water — Frequency Analysis • Ctrl-F: Recurrence Interval; Return Period

Q8. [D5-Q08 • MCQ • Annual exceedance probability to return period]

A design storm has an annual exceedance probability (AEP) of 1%. Its return period T is:

Relevant: $T = \frac{1}{P}$

- (A) 1 yr
- (B) 10 yr
- (C) **100 yr** ✓
- (D) 1000 yr

Answer: (C)

$$T = \frac{1}{P} = \frac{1}{0.01} = 100 \text{ yr.}$$

Handbook: Surface Water — Frequency Analysis • Ctrl-F: Recurrence Interval; Return Period

Q9. [D5-Q09 • MCQ • Risk: probability of at least one exceedance in n years]

What is the probability that a 100-year storm will be equaled or exceeded at least once during a 50-year design life?

Relevant: $P(\text{at least one exceedance in } n \text{ yr}) = 1 - \left(1 - \frac{1}{T}\right)^n$

- (A) 10.0%
- (B) 29.3%
- (C) **39.5%** ✓
- (D) 50.0%

Answer: (C)

$$P = 1 - \left(1 - \frac{1}{100}\right)^{50} = 1 - (0.99)^{50} = 1 - 0.605 = 0.395 \approx 39.5\%.$$

Handbook: Surface Water — Frequency Analysis • Ctrl-F: Risk; Recurrence

Q10. [D5-Q10 • MCQ • Composite (area-weighted) runoff coefficient]

A watershed consists of three sub-areas: 8 acres ($C = 0.30$), 6 acres ($C = 0.70$), and 6 acres ($C = 0.90$). The composite runoff coefficient C_{comp} for use in the Rational Method is most nearly:

Relevant: $C_{\text{comp}} = \frac{\sum C_i A_i}{\sum A_i}$

- (A) 0.52
- (B) **0.60** ✓
- (C) 0.65
- (D) 0.72

Answer: (B)

$$C_{\text{comp}} = \frac{0.30(8) + 0.70(6) + 0.90(6)}{8 + 6 + 6} = \frac{2.4 + 4.2 + 5.4}{20} = \frac{12.0}{20} = 0.60.$$

Handbook: Surface Water — Rational Method • Ctrl-F: Rational Method; Composite

Q11. [D5-Q11 • MCQ • Rational Method units check]

In the USCS Rational Method equation $Q = CIA$, when C is dimensionless, I is in inches per hour, and A is in acres, the resulting Q has units of:

Relevant: $Q = CIA$; $1 \text{ in/hr} \times 1 \text{ acre} \approx 1 \text{ cfs}$

- (A) gallons per minute
- (B) acre-inches per hour
- (C) **ft³/s (cfs)** ✓
- (D) ft/s

Answer: (C)

By the unit conversion $1 \text{ in/hr} \times 1 \text{ acre} = 1.0083 \text{ cfs} \approx 1 \text{ cfs}$, so $Q = CIA$ yields peak flow in ft³/s (cfs) without any unit-conversion factor.

Handbook: Surface Water — Rational Method • Ctrl-F: Rational Method

Q12. [D5-Q12 • MCQ • IDF curve: intensity vs. duration]

On an Intensity–Duration–Frequency (IDF) curve, for a fixed return period (e.g., 100-year), as the storm duration increases from 5 minutes to 24 hours, the design rainfall intensity:

Relevant: IDF: $i = \frac{a}{(t_d + b)^c}$ (typical empirical form)

- (A) increases
- (B) remains constant
- (C) **decreases** ✓
- (D) first increases then decreases

Answer: (C)

For a fixed return period, IDF curves show that intensity decreases as duration increases — a 100-year, 5-min storm is far more intense than a 100-year, 24-hr storm. This is a fundamental property of all IDF curves.

Handbook: Surface Water — Rainfall and Runoff (IDF) • Ctrl-F: IDF; Intensity-Duration-Frequency

Q13. [D5-Q13 • MCQ • SCS maximum potential retention S (SI)]

A watershed has $CN = 75$. Using the SI version of the SCS equation, the maximum potential retention S is most nearly:

Relevant: $S = \frac{25400}{CN} - 254$ (SI, S in mm)

- (A) 63.5 mm
- (B) **84.7 mm** ✓
- (C) 108.0 mm
- (D) 127.0 mm

Answer: (B)

$S = \frac{25400}{75} - 254 = 338.7 - 254 = 84.7$ mm.

Handbook: Surface Water — SCS Runoff • Ctrl-F: Curve Number; SCS

Q14. [D5-Q14 • MCQ • SCS/NRCS runoff depth (SI)]

A watershed with $CN = 80$ receives total storm precipitation $P = 100$ mm. Using the SI SCS method, the direct runoff depth Q is most nearly:

Relevant: $S = \frac{25400}{CN} - 254$ (mm) $Q = \frac{(P - 0.2S)^2}{P + 0.8S}$

- (A) 35.0 mm
- (B) 44.0 mm
- (C) **50.5 mm ✓**
- (D) 63.5 mm

Answer: (C)

$$S = \frac{25400}{80} - 254 = 63.5 \text{ mm}; I_a = 0.2(63.5) = 12.7 \text{ mm. } P > I_a, \text{ so } Q = \frac{(100 - 12.7)^2}{100 + 0.8(63.5)} = \frac{(87.3)^2}{150.8} = \frac{7621}{150.8} = 50.5 \text{ mm.}$$

Handbook: Surface Water — SCS Runoff • Ctrl-F: Curve Number; SCS

Q15. [D5-Q15 • MCQ • Runoff volume from SCS depth (USCS)]

A 500-acre watershed with $CN = 80$ produces a direct runoff depth of $Q = 2.04$ in from a design storm. The total direct runoff volume is most nearly:

Relevant: $V = Q_{\text{depth}} \times A$ (convert depth in to ft: divide by 12)

- (A) 42 ac-ft
- (B) **85 ac-ft ✓**
- (C) 125 ac-ft
- (D) 170 ac-ft

Answer: (B)

$$V = \frac{2.04}{12} \text{ ft} \times 500 \text{ acres} = 0.170 \times 500 = 85 \text{ ac-ft.}$$

Handbook: Surface Water — SCS Runoff • Ctrl-F: Runoff Volume; SCS

Q16. [D5-Q16 • MCQ • Effect of CN on runoff]

For the same storm precipitation P , a watershed with $CN = 90$ will produce _____ direct runoff than one with $CN = 70$.

Relevant: $S = \frac{1000}{CN} - 10$; higher $CN \Rightarrow$ smaller $S \Rightarrow$ less retention

- (A) less
- (B) **more** ✓
- (C) the same
- (D) cannot be determined without knowing P

Answer: (B)

Higher $CN \rightarrow$ smaller $S \rightarrow$ smaller initial abstraction $I_a = 0.2S \rightarrow$ runoff begins sooner and the denominator in the SCS equation is smaller, yielding more runoff Q for the same P .

Handbook: Surface Water — SCS Runoff • Ctrl-F: Curve Number

Q17. [D5-Q17 • MCQ • Runoff coefficient C — physical meaning and range]

In the Rational Method $Q = CIA$, the runoff coefficient C represents the fraction of rainfall that becomes direct runoff. Its physically valid range is:

Relevant: $0 \leq C \leq 1$

- (A) -1 to $+1$
- (B) **0 to 1** ✓
- (C) 0 to 10
- (D) 1 to 100

Answer: (B)

C is a dimensionless ratio (runoff/rainfall); it cannot exceed 1 (no more runoff than rain) and cannot be negative. Typical values: 0.10 (woods) to 0.95 (impervious pavement).

Handbook: Surface Water — Rational Method • Ctrl-F: Runoff Coefficient; Rational

Q18. [D5-Q18 • MCQ • Annual exceedance probability of the 25-year event]

The annual exceedance probability (AEP) of the 25-year design storm is:

Relevant: $AEP = \frac{1}{T}$

- (A) 1%
- (B) 2%
- (C) 4% ✓
- (D) 25%

Answer: (C)

$$AEP = \frac{1}{T} = \frac{1}{25} = 0.04 = 4\%.$$

Handbook: Surface Water — Frequency Analysis • Ctrl-F: Return Period; Recurrence Interval

Q19. [D5-Q19 • Numeric • SCS runoff volume (SI)]

A 200-ha watershed has CN = 75 and receives a total storm precipitation of $P = 120$ mm. Using the SI SCS method, determine the total direct runoff volume (in m^3).

Relevant: $S = \frac{25400}{CN} - 254$ (mm) $Q = \frac{(P - 0.2S)^2}{P + 0.8S}$ (mm) $V = Q \times A$ (convert mm to m, ha to m^2)

Numeric entry.

Answer: $\approx 113,000 m^3$

$$S = \frac{25400}{75} - 254 = 84.7 \text{ mm}; I_a = 0.2(84.7) = 16.9 \text{ mm. } Q = \frac{(120 - 16.9)^2}{120 + 0.8(84.7)} = \frac{(103.1)^2}{187.7} = 56.6 \text{ mm.}$$

$$V = 0.0566 \text{ m} \times 200 \times 10,000 \text{ m}^2 \approx 113,000 \text{ m}^3.$$

Handbook: Surface Water — SCS Runoff • Ctrl-F: Curve Number; SCS

Q20. [D5-Q20 • Numeric • Rational Method peak discharge (USCS)]

A 120-acre residential watershed has a composite runoff coefficient $C = 0.45$ and a 10-year, 1-hour rainfall intensity of 2.8 in/hr. Using the Rational Method, find the peak discharge Q (in cfs).

Relevant: $Q = CIA$

Numeric entry.

Answer: ≈ 151 cfs

$$Q = CIA = 0.45 \times 2.8 \times 120 = 151.2 \approx 151 \text{ cfs.}$$

Handbook: Surface Water — Rational Method • Ctrl-F: Rational Method